



# A bi-subpopulation coevolutionary immune algorithm for multi-objective combinatorial optimization in multi-UAV task allocation

Xi Chen<sup>1</sup> · Yu Wan<sup>1</sup> · Jingtao Qi<sup>1</sup> · Zipeng Zhao<sup>1</sup> · Yirun Ruan<sup>1</sup> · Jun Tang<sup>1</sup>

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## Abstract

With the development of Unmanned Aerial Vehicle (UAV) technology towards multi-UAV and UAV swarm, multi-UAV cooperative task allocation has more and more influence on the success or failure of UAV missions. From the operational research point of view, such problems belong to high-dimensional combinatorial optimization problems, which makes the solving process face many challenges. One is that the discrete and high-dimensional decision variables make the quality of the solution obtained with acceptable time not guaranteed. Second, the desired solution of real missions often needs to satisfy multiple objective functions, or a set of solutions for decision-making. Therefore, this paper constructs a Multi-objective Combinatorial Optimization in Multi-UAV Task Allocation Problem (MCOTAP) model, and proposes a Bi-subpopulation Coevolutionary Immune Algorithm (BCIA). The two coevolutionary mechanisms improve the lower limit of population diversity, and the evolutionary strategy pool integrating multiple strategies and the adaptive strategy selection mechanism enhance the local search ability in the late evolution. In the experiments, BCIA competes fairly with the mainstream multi-objective evolutionary algorithms (MOEAs), multi-objective immune algorithms (MOIAs) and the recently proposed multi-UAV mission planning algorithms. The experimental results on different test problems (including several multi-objective combinatorial optimization benchmark problems and the proposed MCOTAP model) show that BCIA has superior performance in solving multi-objective combinatorial optimization problems (MCOPs). At the same time, the effectiveness of each design component of BCIA has been comprehensively verified in the ablation study.

**Keywords** Coevolutionary algorithm · Multi-objective combinatorial optimization · Multi-objective immune algorithm · Multi-UAV · Task allocation problem

## Introduction

With the advancement of informatization and intelligence in society today, Unmanned Aerial Vehicle (UAV) technology has developed rapidly. Developed from single UAV technology, multi-UAV systems begin to play an important role in applications such as disaster rescue [1], urban security [2], agricultural remote sensing [3], traffic control [4], wireless communication [5], and military operations [6]. Under the influence of these complex application backgrounds, the requirements in multi-UAV missions are more and more diverse, the elements are more and more complex, the constraints are more and more strict, and the scheduling is more

Xi Chen and Yu Wan have contributed equally to this work and should be considered co-first authors.

✉ Jun Tang  
tangjun06@nudt.edu.cn

Xi Chen  
chenxi18@nudt.edu.cn

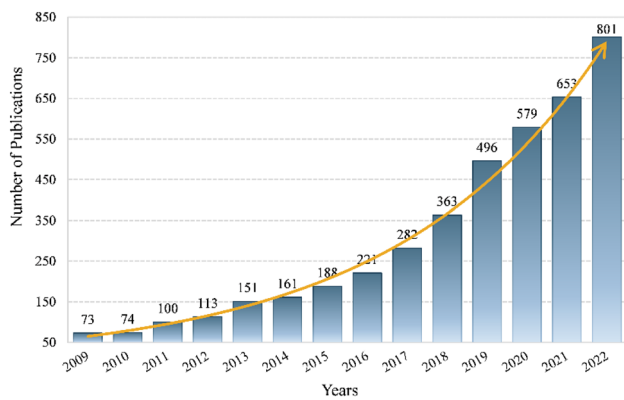
Yu Wan  
wanyu13@nudt.edu.cn

Jingtao Qi  
qijingtao14@nudt.edu.cn

Zipeng Zhao  
zhaozipeng22@nudt.edu.cn

Yirun Ruan  
ruanyirun@nudt.edu.cn

<sup>1</sup> Laboratory for Big Data and Decision, National University of Defense Technology, Changsha 410073, China



**Fig. 1** The number of publications on UAVs task allocation or mission planning included in Web of Science over the years. Note: TS = ((UAV OR UAVs OR (Multi-UAV) OR (Multiple UAVs) OR (Unmanned Aerial Vehicle) OR (Unmanned Aerial Vehicles)) AND ((Task Allocation) OR (Task Assignment) OR (Target Allocation) OR (Target Assignment) OR (Mission Assignment) OR (Task Planning) OR (Mission Planning)))

and more difficult. Therefore, in the process of developing towards intelligence, autonomy and clustering, the first challenge for multi-UAV systems is the collaborative task allocation. Web of Science statistics in Fig. 1 show that during the 15 years from 2009 to the completion of this paper, the related research results (on the topic of UAVs Task Allocation or Mission Planning) has increased year by year. Especially in recent years, the number of papers published in international high-level journals has remained high, and multi-UAV collaborative task allocation has become one of the research hotspots in academia.

In essence, multi-UAV task allocation is to obtain the task execution plan of each UAV according to its type and attributes, and of course, the requirements and number of tasks to be solved urgently, so as to achieve the optimum or satisfying quasi optimum. Special attention should be paid to the constraints brought by the environment or requirements, such as UAV collaboration constraints, task timing constraints, and UAV performance constraints, which should be fully considered in the planning process. These constraints that need to be fully considered in the planning process not only determine the feasibility of scheduling schemes, but also sometimes affect the safety of UAVs. Multi-UAV cooperative task allocation problem belongs to combinatorial optimization problem in Operations Research. The difficulties and challenges of this kind of problem mainly include: (1) the decision variables in these problems are discrete task sequence numbers, which greatly increases the number of infeasible solutions in the solving process. (2) With the expansion of the problem scale, such as the increase of the number of UAVs and tasks, the difficulty of solving increases exponentially, which is often called NP-Hard

(Non-deterministic Polynomial Hard) problem. (3) In practical applications, there is more than one standard to measure the task scheme, so how to balance the multiple costs or objectives of decision-maker brings great challenges to the solution of the problem.

Traditional mathematical models used to formalize the multi-UAV task allocation problem include Multiple Traveling Salesman Problem (MTSP) [7], Vehicle Routing Problem (VRP) [8], Multiple-Choice Knapsack Problem (MCKP) [9], and Mixed-Integer Linear Programming (MILP) [10], etc. But since these models are often transferred from other problems, we need a lot of adaptive improvements. In recent years, due to the improvement of research heat, scholars have proposed the Cooperative Multiple Task Allocation Problem (CMTAP) which is more suitable for UAVs' task execution process [11]. However, including CMTAP, most of the current models only use a single objective as the evaluation criterion, which can no longer meet the increasingly complex actual requirements of multi-UAV task. Multi-objective Optimization Problem (MOP) model can effectively make up for this defect. Multiple optimization objective can provide multi-index evaluation of task allocation schemes, which can provide more comprehensive decision-making information and more diverse allocation schemes for decision-makers. Guang et al. constructed a heterogeneous CMTAP model with the maximum execution time of the multi-UAV system and the cumulative execution time of each UAV as the two objective [12]. Han et al. used the threat cost and the resource loss cost as evaluation indicators of multi-UAV cooperative task allocation model [13]. In [14] and [15], Wang et al. selected the total flight range and the total task completion time as objective functions of the proposed multi-objective UAV cooperative multi-task allocation problem (M-CMTAP).

In order to form a multi-UAV task allocation scheme, it is far from enough to only have the above-mentioned multi-UAV task allocation problem model, and it is also necessary to design an efficient solution algorithm. Traditional mathematical methods often need to calculate gradient when dealing with such MOPs, so the computational time complexity is high, which cannot meet the timeliness requirements of UAVs task allocation. In recent years, Multi-Objective Evolutionary Algorithms (MOEAs) have performed well in solving numerical optimization problems, and have become a research hotspot. The representative MOEAs include Nondominated Sorting Genetic Algorithm (NSGA) [16], NSGA-II [17], Strength Pareto Evolutionary Algorithm 2 (SPEA2) [18], Multi-Objective Evolutionary Algorithm based on Decomposition (MOEA/D) [19], Multi-Objective Particle Swarm Optimization (MOPSO) [20], etc. Their essential core elements are population evolution mechanisms such as survival of the fittest and genetic mutation. Although they cannot guarantee the exact global optimum, they are superior in search ability, simplicity, efficiency,

robustness and so on. Therefore, many scholars have tried to improve the coding methods and evolutionary mechanisms of the multi-objective numerical optimization algorithm, and formed discrete multi-objective optimization algorithms for UAVs task allocation. For example, Wang et al. [14] adopted a hybrid discrete coding method based on task allocation and path planning, and introduced the competitive coordination mechanism into MOPSO to solve the M-CMTAP model. Xu et al. [21] integrated variable neighborhood search strategy and feasibility recovery strategy into NSGAI algorithm, which effectively solved the task allocation problem for multi-vehicle systems and showed high robustness. Ma et al. [22] combined NSGA-II with multi-task optimization (MTO) to solve the task planning problem of multi-UAV fast maritime search and rescue (MSR) with three objective functions. Inspired by the artificial immune system, Multi-Objective Immune Algorithms (MOIAs) are in the ascendant. These algorithms have become popular choices for solving MOPs due to their advantages of better convergence and higher search efficiency [23]. Clonal Selection Algorithm (CLONALG) [24] and Nondominated Neighbor Immune Algorithm (NNIA) [25] are typical representatives and many studies such as Adaptive Immune-inspired Multi-objective Algorithm (AIMA) [23] have also been carried out on this basis. However, in general, there are few studies on MOIAs for Multi-objective Combinatorial Optimization Problems (MCOPs), and the application of MOIAs in multi-UAV task allocation is even more lacking.

In this paper, the multi-UAV task allocation problem is abstracted as a general MCOP. Based on the CMTAP model, we construct a Multi-objective Combinatorial Optimization in Multi-UAV Task Allocation Problem (MCOTAP). As for the algorithm to solve MCOTAP, we innovatively propose Bi-subpopulation Coevolutionary Immune Algorithm (BCIA) inspired by NNIA's idea of focused search near dominant individuals and AIMA's idea of adaptively selecting different evolutionary strategies. BCIA establishes the search mechanism of bi-subpopulation co-evolution, in which offspring individuals are generated by adaptive cooperative evolution within the elite subpopulation or by adaptive guided evolution between the elite subpopulation and the common subpopulation. This mechanism provides a guarantee for the directionality of the search and the diversity of the population. In addition, we design three evolutionary strategies for discrete variables based on classical crossover operators. BCIA adaptively selects different evolutionary strategies according to the evolutionary stage. Therefore, BCIA has better population diversity in the early evolutionary stage and faster convergence speed in the late evolutionary stage. The main contributions of BCIA are as follows.

- By introducing the bi-subpopulation coevolution mechanism, BCIA uses the collaboration between elite antibodies and the guidance of elite antibodies to common antibodies to co-generate the offspring antibody population. This mechanism increases the diversity of offspring antibody population. Therefore, the drawback of traditional immune algorithms that it is easy to converge prematurely to the local optimum is also avoided.
- By introducing the adaptive strategy selection mechanism, BCIA can choose different evolutionary strategies with different features (i.e., tendency of diversity or convergence) according to the different stages of population evolution. Therefore, BCIA has better population diversity in the early evolutionary stage and faster convergence speed in the late evolutionary stage.
- By improving several existing crossover operators, BCIA forms an evolutionary strategy pool. These evolutionary strategies have the adaptability of MCOPs and the selectivity in evolution, so they can better guide the evolutionary direction.
- BCIA is a new attempt to apply the improved MOIA to solve the MCOPs. At the same time, it has also unprecedentedly applied to the practical problem of multi-UAV task assignment.

This paper is organized as follows: Sect. "[Problem formulation](#)" describes the core essence of the multi-UAV task allocation problem and common application scenarios, and subsequently constructs the MCOTAP model. In Sect. "[The proposed BCIA](#)", BCIA, the highlight of this paper, is explained in detail. Firstly, the contribution of BCIA is clearly summarized, and then the overall design idea is expounded. Finally, the details and functions of each core module are introduced. The simulation experiments section is in Sect. "[Experimental study](#)". We compare BCIA with state-of-the-art MOEAs and MOIAs on both the multi-objective combinatorial optimization benchmark problems and the proposed MCOTAP. Combined with the results of comparative experiments, the analysis of the advancement of BCIA followed. In addition, we establish an ablation study to verify the effectiveness of the proposed mechanism and operators in BCIA. Finally, Sect. "[Conclusions and future work](#)" concludes the work and discusses future research directions.

## Problem formulation

In missions such as urban security, international counterterrorism, disaster rescue or military operations, it is usually necessary to reconnoitre some key targets. UAVs with collaborative capability is the best choice for such reconnaissance missions due to their advantages of low cost, fast response and high fault tolerance [26]. Before the mission begins,

**Table 1** Summary of attributes and corresponding symbols of UAVs and targets

| Attributes                       | Symbols |
|----------------------------------|---------|
| UAV                              | $UAV_i$ |
| Number of UAVs                   | $N$     |
| Velocity of UAVs                 | $Vol$   |
| Maximum endurance                | $Time$  |
| Take-off point                   | $S$     |
| Target                           | $T_j$   |
| Number of targets                | $N_T$   |
| Coordinate of targets            | $Tp_j$  |
| Reconnaissance duration required | $ToR_j$ |
| Reconnaissance task for $T_j$    | $M_j$   |
| Temporal relations between tasks | $TC$    |

we need to develop a mission execution plan for each UAV based on the tasks, objectives, constraints, and evaluation criteria. This is exactly the practical problem to be solved in this paper: the task allocation of multi-UAVs in multi-target reconnaissance missions. We abstract the multi-UAV task allocation problem as a general combinatorial optimization problem, and further construct the MCOTAP model based on the CMTAP. The MCOTAP model has several characteristics compared with the existing multi-UAV task allocation problem model. It maps all the elements of multi-UAV task allocation to MCOPs, so many multi-objective combinatorial optimization algorithms can be directly applied to solve MCOTAP only by adjusting the encoding of variables. The constraint setting of MCOTAP is more reasonable. Instead of adding all constraints to the model, it discards constraints that are not part of the task allocation consideration. Moreover, it formulates the encoding in a way that allows the UAV to be idle. This is forgotten by most models of the multi-UAV task allocation problem.

MCOTAP can be represented as the quadruple tuple  $\{MUAS, Targets, Constraints, Functions\}$ . Among them,  $MUAS$  is the Multi-UAV System, which contains all  $N$  UAVs and their attributes such as cruising velocity, maximum endurance and take-off point.  $Targets$  represent all  $N_T$  targets in the reconnaissance mission, along with attributes such as their coordinates and the reconnaissance duration required. Table 1 summarizes the detailed attributes of UAVs and targets and their corresponding symbolic representations.

*Constraints* represent the constraints or requirements on the task assignment plan (i.e., the solution to the problem) according to the task background or target attributes. There are the three main constraints in MCOTAP.

(1) UAV endurance constraint: whether powered by battery or fuel, the endurance of UAVs is constrained by battery capacity or combustible fuel load. Therefore, we uniformly

express this constraint in terms of maximum endurance time. Let  $Time_i^{all}$  be the total mission time of  $UAV_i$ , then the endurance constraint can be expressed as the following equation.

$$Time_i^{all} \leq Time \quad (1)$$

(2) UAV velocity constraint: Each type of UAV has a limited maximum flight velocity due to limitations such as UAV performance and air disturbance. Therefore, the cruising velocity of UAVs during the mission cannot exceed the maximum flight velocity. Let  $Vol_{ty}^{max}$  denotes the maximum velocity of UAVs of type  $ty$ , then the UAV velocity constraint can be expressed as follows.

$$Vol_{ty} < Vol_{ty}^{max} \quad (2)$$

(3) UAV temporal relation constraint: In a specific mission scenario, different reconnaissance targets have different importance, so there are temporal relations between tasks. We set the temporal relation matrix  $TC = (tc_{ab})_{N_T \times N_T}$ , where  $tc_{ab} = 1$  indicates that the execution priority of  $M_a$  (reconnaissance task of target  $T_a$ ) is higher than  $M_b$  (reconnaissance task of target  $T_b$ ), otherwise  $tc_{ba} = 1$ . If  $tc_{ab} = tc_{ba} = 0$ , it means that there is no temporal relation constraint between  $M_a$  and  $M_b$ . Then the formal representation of the UAV temporal relation constraint of MCOTAP is as follows.

$$\forall a, b \in [1, N_T], tc_{ab} = 1 \prec Time_a^{finish} \leq Time_b^{finish} \quad (3)$$

where  $Time_a^{finish}$  and  $Time_b^{finish}$  denote the time of completion of  $M_a$  and  $M_b$  according to the task assignment scheme. The ' $\prec$ ' belongs to the modal logic notation and denotes strict implication.

*Functions* represents the multiple objectives of MCOTAP, which is the standard we use to evaluate the quality of solutions. In the multi-UAV reconnaissance mission, we generally hold that the faster the task is completed and the less the resource is consumed, the better the task allocation scheme is. The total completion time is the time taken by the last UAV to complete tasks, including flight time and reconnaissance time. Resource consumption can be expressed by the total flight range of all UAVs performing the mission. Let's consider the scenario where the distance between the take-off point and the mission area is much greater than the distance between the targets, the fewer UAVs perform the mission, the total flight range we consume. But obviously, the total completion time will increase due to the increase of reconnaissance targets by single UAV. Therefore, these two objective functions are generally in conflict. The mathematical formula describing these two objective functions is as

follows.

$$\begin{cases} \text{Min} : F_{\text{time}} = \max_{1 \leq i \leq N} \text{Time}_i^{\text{all}} \\ \text{Min} : F_{\text{flight}} = \sum_{i=1}^N \sum_{m=1}^{N^i-1} \text{Flight}(Tp_m^i, Tp_{m+1}^i) \end{cases} \quad (4)$$

where  $N^i$  denotes the number of tasks of  $UAV_i$ ,  $\text{Flight}(Tp_m^i, Tp_{m+1}^i)$  denotes the flight range from the  $m$ th task target to the  $m+1$ st task target in the task allocation scheme of  $UAV_i$ .

In addition to the above elements, another issue that has attracted much attention in task allocation is the encoding of variables. In most of the current research on multi-UAV task allocation, whether the decision variables are discrete or continuous or both, there is a common flaw that all UAVs must participate in the mission. But when put into real-world missions, this is clearly unscientific and unreasonable. Sometimes, too many UAVs involved in a mission cannot significantly improve the task completion time, but will greatly increase the whole resource consumption. Take the case of three UAVs working together on a mission, for example. If the mission target is located behind enemy lines, meaning that the distance between the command center and the target is great, while the target is relatively close and the time required for reconnaissance of a single target is not long. In this case, sending three UAVs to complete the mission compared to sending two UAVs will only slightly shorten the overall mission time (the time taken by the UAV that completes the mission last), but will greatly increase resource consumption because one additional UAV has to fly back and forth.

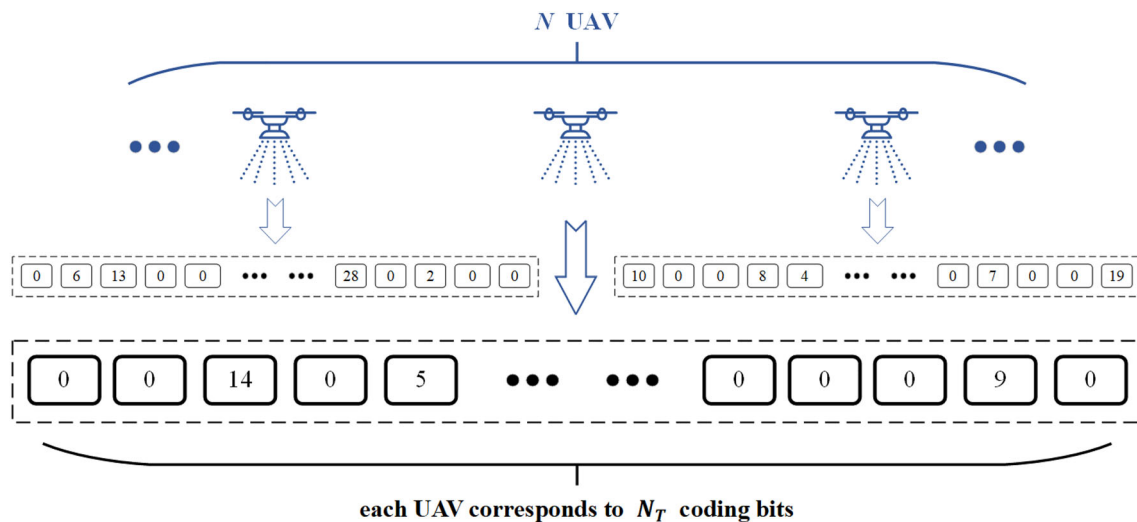
Therefore, this paper innovatively proposes a new hybrid encoding method. As shown in Fig. 2, in a multi-UAV system with  $N$  UAVs, each UAV corresponds to  $N_T$  coding bits. The default value for these is 0. If the value of a coding bit is  $x \in [1, N_T]$ , it means that the corresponding UAV performs a reconnaissance task on target  $x$ . In this way, the other coding bits except 0 constitute the task allocation scheme and execution sequence of this UAV. A total of  $N \times N_T$  coding bits for all  $N$  UAVs together form the complete encoding of multi-UAV system. A few special cases deserve special mention: (1) The situation of multiple UAVs executing the same task is mapped into the encoding as follows: the same task coding exists in the task sequence of two or more UAVs simultaneously. (2) If a single UAV executes the same task multiple times, the same task coding is merged, and the task sequence number only appears once in the UAV's coding. This is because if the UAV interrupts a task to perform another task, and then returns to perform the same task later, it will cause an increase in flight range. (3) If the coding bits of one UAV is all 0, it means that this UAV does not participate in mission but will get a higher cost efficiency ratio.

## The proposed BCIA

This section introduces the Bi-subpopulation Coevolutionary Immune Algorithm for multi-objective combinatorial optimization in multi-UAV task allocation, which is called BCIA. Compared with MOEAs, MOIAs has the characteristics of high search efficiency and good convergence, which is naturally related to the problem of task allocation which requires efficiency. However, this kind of algorithm needs to solve the defects of poor diversity and easy to fall into local optimum. Therefore, BCIA introduced a bi-subpopulation coevolution mechanism and an adaptive strategy selection mechanism, and established an evolutionary strategy pool containing multiple crossover operators. These mechanisms and operators enable BCIA to better balance diversity and convergence.

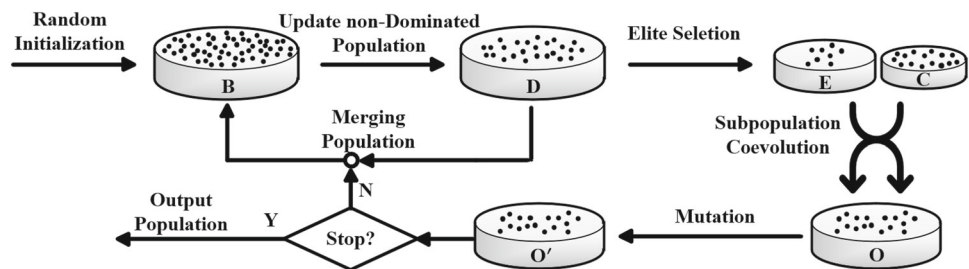
The population evolution of BCIA is shown in Fig. 3. First, the antibody population  $B$  is randomly initialized. The criterion of initialization is that the encodings of antibody individuals satisfy the constraints, which improves the effectiveness of initial antibodies. By Pareto non-dominated sorting,  $B$  is screened to obtain dominant antibody population  $D$ . Furthermore, the elite antibody population  $E$  and common antibody population  $C$  were further formed according to the crowding-distance. In general,  $E$  consists of antibodies on the Pareto-optimal front. However, when the individuals on the Pareto-optimal front is too few, the individuals on the next front are selected to make up until the elite population size reaches  $nE$ . It is worth noting that the size of population  $E$  is smaller than both  $D$  and  $C$ , that is,  $nE < nC < nD$ .

Subsequently, BCIA applied an improved clonal selection to the elite antibody population  $E$ , which is our adaptive coevolution. According to the selection idea in CLONALG [24], the elite antibodies in  $E$  are the selected antibodies with best affinity. Instead of following the original Cloning procedure to directly replicate these antibodies in proportion, we defined a rating  $r$  for each elite antibody, which measures how excellent the antibody is. An elite antibody with a larger  $r$  will have more coevolution times and generate more offspring antibodies accordingly. The adaptive coevolution can be divided into two cases, where elite antibodies follow a certain probability to select elite antibodies in the same  $E$  to perform cooperative evolution or select common antibodies in  $C$  to perform guided evolution. The former emphasizes more on fair competition between two elite antibodies, while the latter emphasizes on guiding the evolution of ordinary antibodies toward elite antibodies. BCIA fuzzy divides the population evolution into the early stage and the late stage. During the coevolution, BCIA adaptively selects different strategies from the evolutionary strategy pool according to the different requirements of early stage or late stage for population diversity and convergence. The above adaptive coevolution generates offspring antibody population  $O$ . Finally, BCIA merges the antibody population  $O'$  formed after mutation



**Fig. 2** An example of the new hybrid encoding method for multi-UAV task allocation problem

**Fig. 3** The general framework of the population evolution process of BCIA



with the dominant antibody population  $D$  to form the next generation population  $B$ , and entered a new round of evolution.

To show the details of BCIA, each key module is specially explained next.

### Update dominant population

The Update Dominant Population module is used to perform a preliminary screening of antibody population  $B$ . The better  $nD$  antibodies are obtained to form the dominant antibody population  $D$ . The algorithmic steps of this module refer to the fast non-dominated sorting approach in [17]. Firstly, according to the affinity, we sort the antibodies in  $B$  by non-dominated sorting and construct the non-dominated set *FrontNo*. That is, to obtain the ordinal numbers of Pareto fronts where each antibody is located. Subsequently, *CrowDis* i.e., the crowding distance of antibodies on each front is calculated. Then according to *FrontNo*, *CrowDis* and  $nD$ , we can get which antibodies in  $B$  can enter  $D$ .

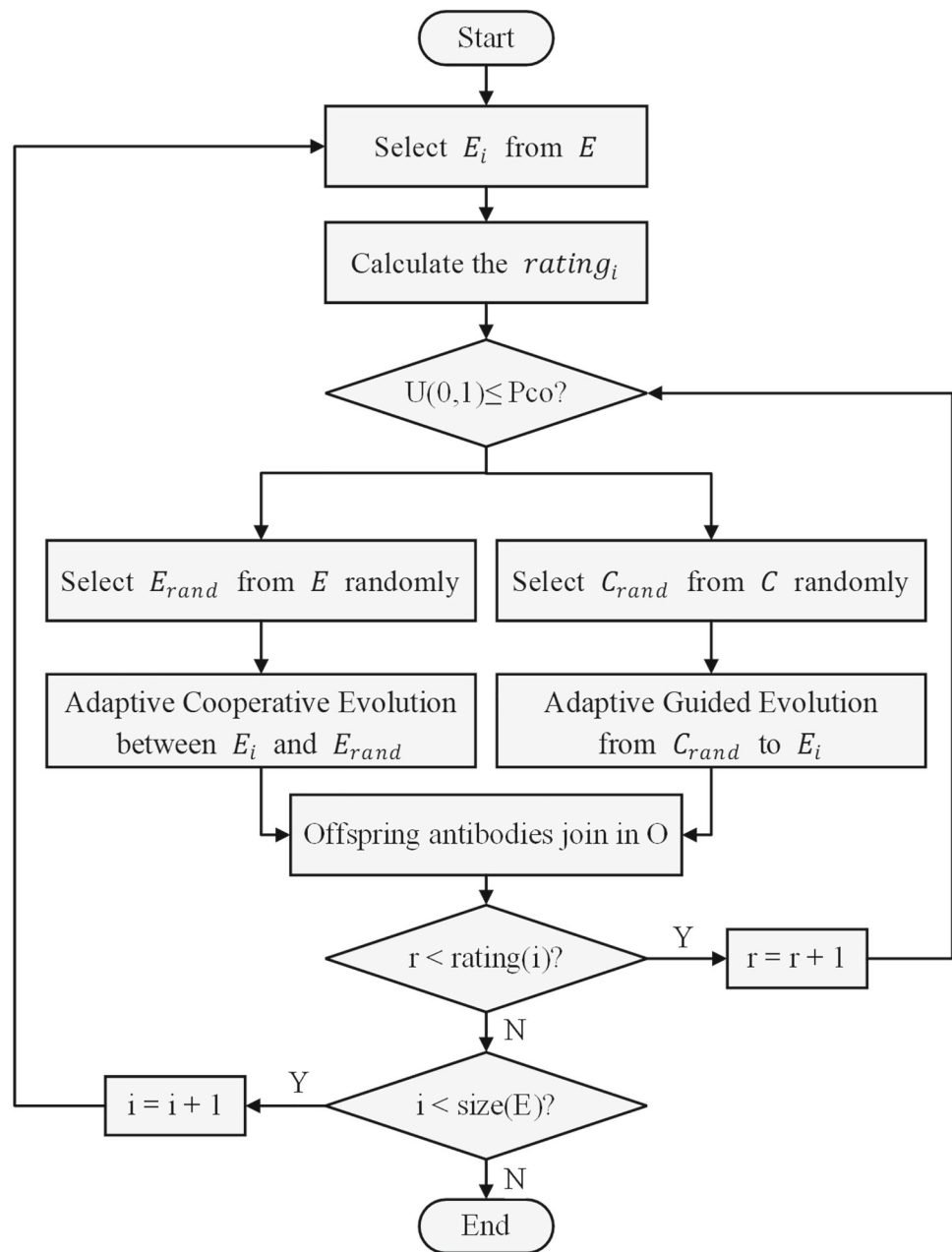
### Elite selection

The function of the Elite Selection module is to further screen the dominant population  $D$  to select the best antibodies that

are called elite antibodies. The selected elite antibodies form the elite antibody population  $E$ . Different from the traditional immune algorithm based on clonal selection, we set the population size  $nE$ . When the antibodies on Pareto-optimal front of  $D$  are insufficient, the individuals with sparse crowding-distance are selected from the next Pareto front to complement  $E$ . This initiative can improve the shortcomings of original clonal selection methods, i.e., easy to fall into search stagnation and lack of population diversity. The remaining antibodies in  $D$  form the common antibody population  $C$ , which is used for adaptive guided evolution with  $E$  in the next module. The algorithm steps shown in Table 2 are similar to Update Dominant Population module.

### Subpopulation coevolution

Subpopulation Coevolution is the most critical module of BCIA. It defines the way populations evolve and largely determines the efficiency of the algorithm. Figure 4 is the flowchart of the entire module. The inputs of this module, which are the two subpopulations involved in coevolution, is the Elite antibody population  $E$  and the common antibody population  $C$ . Each elite antibody in  $E$  is selected in turn to perform coevolution. We assign a  $rating_i$  to each elite antibody  $E_i$ , which determines how many times  $E_i$  performs

**Fig. 4** Flowchart of the Subpopulation Coevolution Module

coevolution and therefore how many offspring antibodies are generated based on  $E_i$ . Subsequently following the probabilistic  $P_{co}$ ,  $E_i$  crosses with a random individual  $E_{rand}$  from  $E$  to produce offspring. The choice of crossover operators is based on the current evolutionary stage. Because the crossover parents are all from  $E$ , we call this evolution Adaptive Cooperative Evolution. In the other case ( $U(0,1) > P_{co}$ ), we randomly select antibodies  $C_{rand}$  from the common antibody population  $C$  to cross with  $E_i$ . In this case, the crossover operators can retain more information about  $E_i$ , and the purpose is to guide the common antibody to evolve in the direction of the elite antibody, that

is, in the direction of improving the antibody affinity. We define this evolution as Adaptive Guided Evolution. The key technologies of Subpopulation Coevolution module, such as Elite Antibody Rating, Evolutionary Strategy Pool, Adaptive Cooperative Evolution, and Adaptive Guided Evolution, will be detailed later. It should be noted in particular that the population  $O$  generated by the Subpopulation Coevolution module is not the final offspring population. In order to improve the diversity of BCIA, we also use the Slight Mutation operator to deal with  $O$ . Specifically, a single mutated bit is randomly selected for all individuals in  $O$  and swapped with the codes at other positions. These individuals constitute

**Table 2** Algorithm steps of the elite selection module

|                                                                                                         |                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Input:</b> Dominant antibody population $D$ , Desired size $nE$ of the Elite antibody population $E$ |                                                                                                                                                                                  |
| <b>Output:</b> The Elite antibody population $E$ , The Common antibody population $C$                   |                                                                                                                                                                                  |
| 1                                                                                                       | Non-dominated Sorting: sort the antibodies in $D$ and construct the non-dominated set $FrontNo$                                                                                  |
| 2                                                                                                       | Calculate the crowding distance: $CrowDis$ i.e., the crowding distance between antibodies in $D$ is calculated front-by-front                                                    |
| 3                                                                                                       | The largest front sequence number that can be fully accommodated within the scale $nE$ is calculated according to $FrontNo$ , which is denoted as $lastFNo$                      |
| 4                                                                                                       | Record the index of the antibodies in $D$ from front 1 to front $lastFNo$ , denoted as $Index$                                                                                   |
| 5                                                                                                       | The sequence number of the next front from which the antibodies will be selected is $nextFNo = lastFNo + 1$ , and the number of needed antibodies is $nextNum = nE - sum(Index)$ |
| 6                                                                                                       | The antibodies on $nextFNo$ are sorted in descending order according to $CrowDis$ , and the index of the first $nextNum$ antibodies is added into $Index$                        |
| 7                                                                                                       | Obtain the Elite antibody population $E = D(Index)$ and the Common antibody population $C = D - E$                                                                               |
| 8                                                                                                       | <b>return</b> $E, C$                                                                                                                                                             |

the final offspring population  $O'$  and enter the next round of evolution.

### Elite antibody rating

Elite Antibody Rating is a measure of how good an elite antibody is and determines the number of offspring antibodies will be generated based on that elite antibody. Therefore, this value is related to the Pareto dominance relationship between antibodies and the crowding-distance, and is also related to the defined offspring population size. In this algorithm, we define the rating of elite antibody  $E_i$  is as follows.

$$rating_i = \lceil \frac{n_O \times CrowDis_i}{\sum_{m=1}^{n_E} CrowDis_m} \rceil \quad (5)$$

where  $n_O$  represents the expected size of the offspring population,  $n_E$  represents the size of  $E$ , and the  $CrowDis_i$  represents the crowding-distance of  $E_i$ . If  $CrowDis_i = INF$ , we replace INF by twice the maximum value in  $E$  except INF. It should be noted that BCIA sets up a safeguard mechanism for the size of elite population  $E$ , that is, if there are insufficient antibodies on the Pareto-optimal front, antibodies will be added from the next front to complement  $E$ . Therefore, if an antibody does not part of the Pareto-optimal front, the crowding-distance of it will be scaled before calculating rating. The specific calculative formula is as follows

$$CrowDis_i = realCrowDis_i \times 10^{-(FrontNo_i-1)} \quad (6)$$

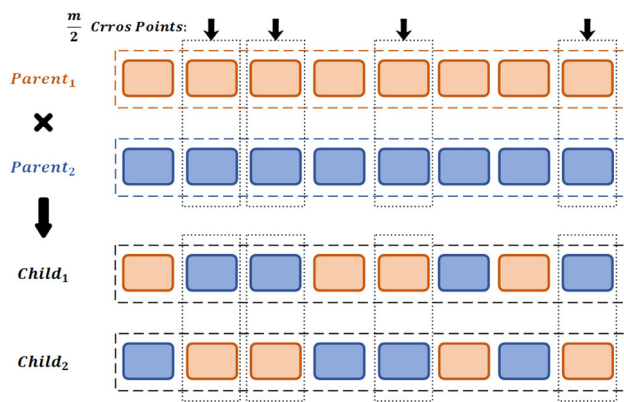
where  $realCrowDis_i$  denotes the true crowding-distance of  $E_i$ , and  $FrontNo_i$  represents the sequence number of the front on which  $E_i$  is located.

### Evolutionary strategy pool

BCIA introduces three crossover operators suitable for MCOPs to form Evolutionary Strategy Pool. The following three evolution strategies with different characteristics are explained in detail. Let  $Parent_1 = \{x_1, x_2, \dots, x_m\}$  and  $Parent_2 = \{y_1, y_2, \dots, y_m\}$  denote the two parent antibodies involved in the evolution, where  $m$  is the length of coding.

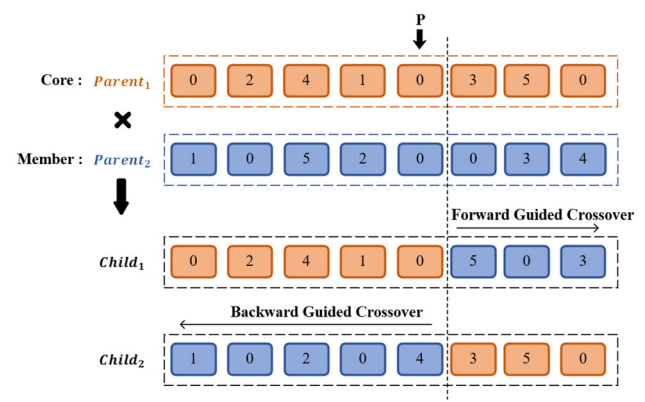
The first evolutionary strategy  $ES_1$  is the Uniform Crossover operator, as shown in Fig. 5. It randomly generates  $m/2$  cross point bits. The parent antibodies then cross over at these bits, that is, interchange the codes.  $ES_1$  produces two offspring antibodies at a time and the information of offsprings comes from both parents uniformly.

The second evolutionary strategy  $ES_2$  is the Random Fragment Crossover operator, whose principle is shown in Fig. 6. First, we define the crossover ratio  $k$ , which is the length of the coding fragment to be crossed. The starting

Fig. 5 Illustration of the evolutionary strategies  $ES_1$ 

point of the fragment  $CrossP_1$  and the symbol (+ or -) of  $k$  are randomly generated. Then we can calculate the  $CrossP_2 = CrossP_1 \pm k$ .  $Parent_1$  and  $Parent_2$  interchange the coding fragment  $\{CrossP_1, CrossP_2\}$  to obtain offspring antibodies  $Child_1$  and  $Child_2$ . There are two aspects to pay special attention to when using. One is the out-of-bounds handling when is generating  $CrossP_2$ . This is already reflected in Fig. 6. The other is that the setting of  $k$  needs to consider the evolutionary demand. The larger the  $k$  is, the stronger the global search ability of  $ES_2$  is. Conversely, a small  $k$  corresponds to a local focus search.

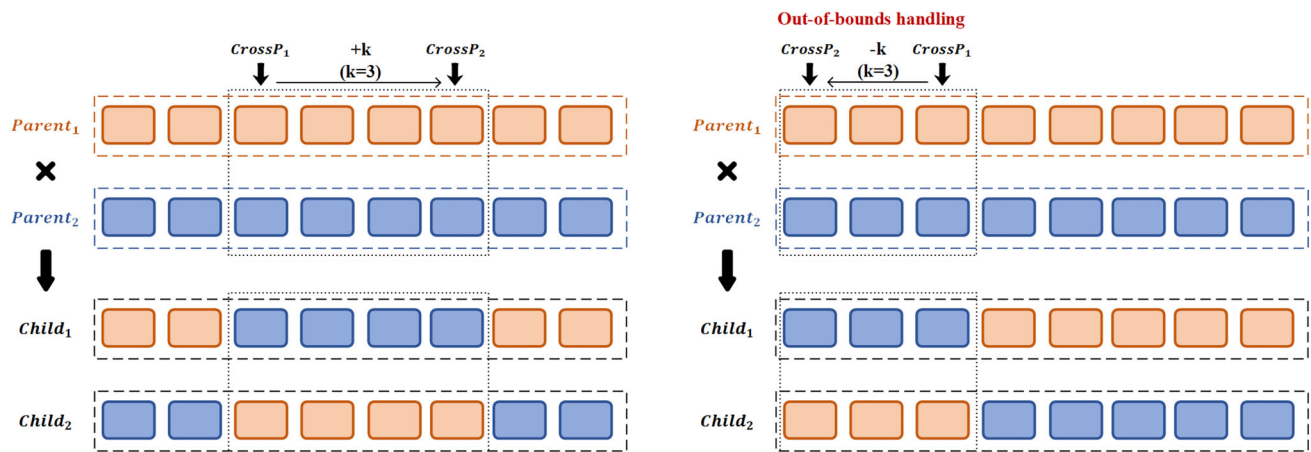
The third evolutionary strategy  $ES_3$  is improved from the Order-Crossover operator, which we call the Bidirectionally Guided Crossover operator.  $ES_3$  will designate one parent individual as the Core of evolution, and the other parent individual as Member. As shown in Fig. 7,  $Parent_1$  is the Core and  $Parent_2$  is the Member. The cross point  $P$  is randomly generated. The Forward Guided Crossover generates the offspring  $Child_1$ , which retains the complete information of Core at the same location up to point  $P$ . The rest information of  $Child_1$  is provided by Member (i.e.,  $Parent_2$ ). It

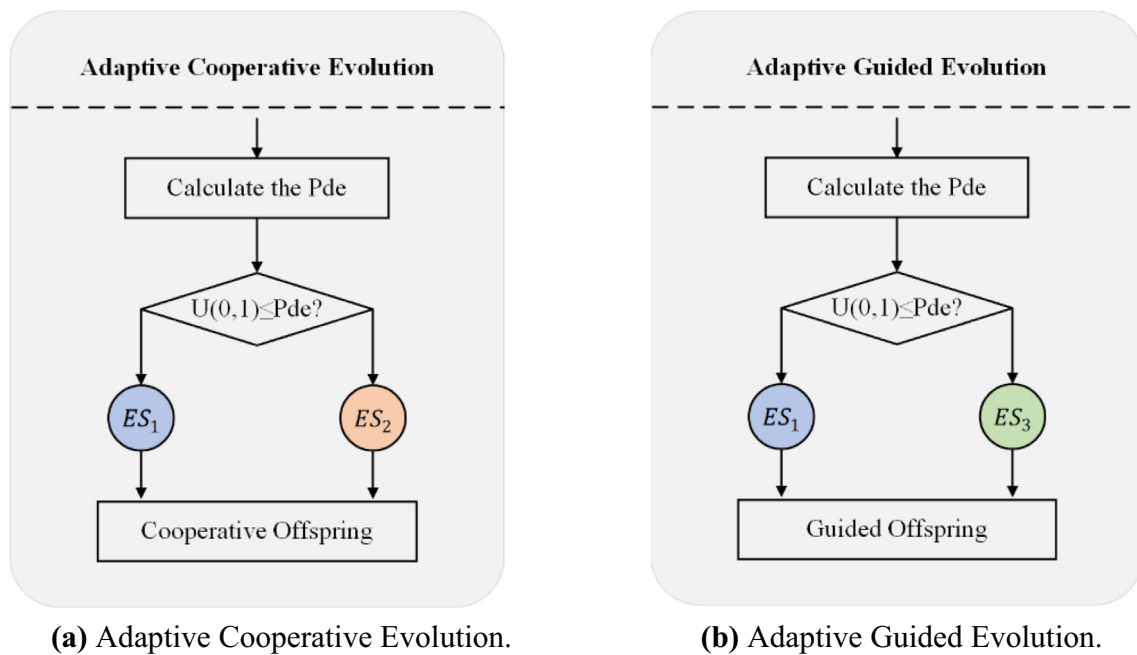
Fig. 7 Illustration of the evolutionary strategies  $ES_3$ 

should be specially noted that in MCOPs with permutation variables, such as task assignment problem, TSP, and vehicle routing problems, the information provided by the Member should be filtered. That is to remove the encodings of Member repeated with the retained coding fragment of Core. This measure can avoid the repeated encoding to produce illegal offspring individuals, so as to improve the efficiency of evolution. The Backward Guided Crossover is similar, but in the opposite direction. It keeps the complete information of Core after  $P$  and generates the offspring individual  $Child_2$ .

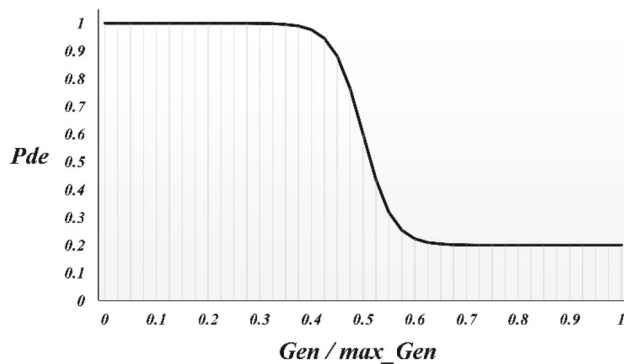
### Adaptive cooperative evolution and adaptive guided evolution

Different stages of evolution have different requirements for algorithm. In the early stage, we should pay attention to the global search ability and pursue the diversity of solutions. In the later stage, we should pay attention to the local search ability and refine the search for better solutions [27, 28]. Therefore, BCIA sets up an adaptive strategy selection mechanism, which selects the corresponding evolution strategies

Fig. 6 Illustration of the evolutionary strategies  $ES_2$



**Fig. 8** Flowcharts of  $E_i$  with  $E_{rand}$  for adaptive cooperative evolution and  $E_i$  with  $C_{rand}$  for adaptive guided evolution



**Fig. 9** The tendency of  $Pde$  as the evolutionary generation increases

from the Evolutionary Strategy Pool according to the current stage. Figure 8a, b show the processes of  $E_i$  with  $E_{rand}$  for Adaptive Cooperative Evolution and  $E_i$  with  $C_{rand}$  for Adaptive Guided Evolution, respectively.

In view of the fact that it is difficult to accurately divide the evolutionary stages, this paper uses a probabilistic approach to divide them into early stage and later stage [23]. Let  $Pde$  denote the execution probability of  $ES_1$ . If  $U(0,1) \leq Pde$ , the evolution is performed following strategy  $ES_1$ . Otherwise, other evolution strategy  $ES_2$  or  $ES_3$  is executed. The specific calculation formula of  $Pde$  is as follows.

$$Pde = 0.2 + (0.8 \times \frac{1}{1 + \exp(35 \times (\frac{Gen}{max\_Gen} - 0.5))}) \quad (7)$$

where  $Gen$  is the current evolutionary generation,  $max\_Gen$  is the maximum evolutionary generation. Figure 9 depicts the tendency of  $Pde$  as the evolutionary generation increases. In the early stage,  $Pde$  tends to 1, indicating that BCIA almost completely adopts  $ES_1$  with stronger global search ability. Therefore, it will produce more differentiated offspring antibodies and increase the population diversity. In the later stage,  $Pde$  approaches 0.2, which means the execution probability of  $ES_2$  or  $ES_3$  is as high as 80%. In this case, the offspring are mostly generated by local search near the current optimum.

## Experimental study

In this section, we compare the proposed BCIA with five classical and excellent multi-objective algorithms to verify the comprehensive performance of BCIA. These excellent algorithms that often appear in comparative studies include NSGA-II, SPEA2, MOEA/D, NNIA and Non-dominated Neighbor Coevolutionary Algorithm (NNCA) [29]. Among them, NNIA is a typical representative of the multi-objective immune algorithm, and NNCA is a typical representative of the multi-objective coevolutionary algorithm. In addition to the simulation experiments and comparison on the proposed MCOTAP, we also select four most classical and frequently used MCOPs as benchmark problems to evaluate the algorithm performance. These benchmark problems include the Multi-Objective Next Release Problem (MONRP) [30], ZDT5 [31], the Multi-Objective Traveling Salesman Problem

(MOTSP) [32] and the Multi-Objective Quadratic Assignment Problem (mQAP) [33], each of which has different characteristics. For example, MONRP and ZDT5 are MCOPs with binary variables, while MOTSP and mQAP are MCOPs with permutation variables. In this way, we can not only test the performance of BCIA in MCOTAP, but also examine the comprehensive ability of BCIA in dealing with MCOPs with different characteristics.

## Experimental setup

All comparative experiments in this section were conducted in the same experimental environment. The details of the hardware and software configurations are attached. CPU: Intel® Xeon® Silver 4210R (13.75 M cache, 2.40 GHz). RAM: 62G. Operating system: Ubuntu 18.04 LTS. Simulation platform: MATLAB R2021a, PlatEMO v4.1 [34].

In this paper, the parameters of the selected comparison algorithms are set in accordance with the source reference. Meanwhile, for fairness, the initial population size of each algorithm is set to 500, the dominant population size and the evolutionary population size are set to 100, and the active antibody population size in the immune algorithm is set to 20. In addition, the elite population size of NNCA is set at 10 according to reference [29], and the proposed BCIA is consistent with it.

Because the MOPs to be solved are discrete, the operators of each comparison algorithms need to be specially explained. The Crossover and Mutation operators of NSGA-II, MOEA/D, SPEA2 and NNIA are provided by PlatEMO platform [34]. They all use Uniform Crossover and Bit-flip Mutation to solve the MCOPs with binary variables and use Order Crossover and Slight Mutation to solve the MCOPs with permutation variables. As for NNCA, we refer to the author's strategy when applying it to MCOPs: the crossover operator is selected from Uniform Crossover and Two-points Crossover with equal probability, and the mutation operator is Slight Mutation [35].

In order to improve the experimental accuracy and reliability, we set the maximum number of function evaluations to 10,000 for all algorithms, and all experiments were run 30 times. Therefore, all experimental results in this section are the average of 30 experiments.

## Experimental results on benchmark problems

This section tests the performance of BCIA on benchmark MCOPs. Most of them are abstracted from real applications in production and life, so they have different characteristics. Their real Pareto-optimal front and the reference point of Hypervolume (HV) are provided by the simulation platform PlatEMO [34]. For quantitative comparison, Table 3 summarizes the HV results of all the considered algorithms on these

benchmark MCOPs. In order to examine whether the results are significantly better/worse, we used the non-parametric Wilcoxon's rank-sum two-sided comparison test at the 0.05 significance level [36]. So, the HV data of the comparison algorithm are marked with "+", "-", and "≈", indicating that the corresponding algorithm is better than, worse than, and close to BCIA, respectively. And the last row in Table 3 counts the number of "+", "-", and "≈". The data filled with dark orange indicates that the algorithm is optimal on the corresponding problem, and the light orange indicates the sub-optimal.

Based on these results, it is clear that BCIA performs best or second best in all benchmark problems. In detail, the comprehensive performance of BCIA is significantly better than other algorithms on ZDT5 and mQAP. BCIA and NNCA have similar performance on MOTSP, and their HV values are significantly better than the other algorithms. Only on the MONRP, BCIA is in second place. As shown in the last row of Table 3, BCIA respectively outperforms NSGA-II, MOEA/D, SPEA2, NNIA, and NNCA on 4, 4, 4, 3, and 3 out of 4 benchmark test MCOPs.

In order to compare the algorithms more intuitively, we take the ZDT5 problem as an example and plot the Pareto front obtained by all algorithms in Fig. 10. Among them, the red solid dots represent the true Pareto-optimal front of the ZDT5 problem [31, 34], the blue solid dots represent the Pareto front obtained by BCIA, and the hollow dots in different colors represent the Pareto front of comparison algorithms. It is easy to find that BCIA has obvious advantages over other algorithms, because the solutions obtained by BCIA are more evenly distributed and closer to the true Pareto-optimal front.

In summary, BCIA significantly outperforms other comparison algorithms on benchmark MCOPs.

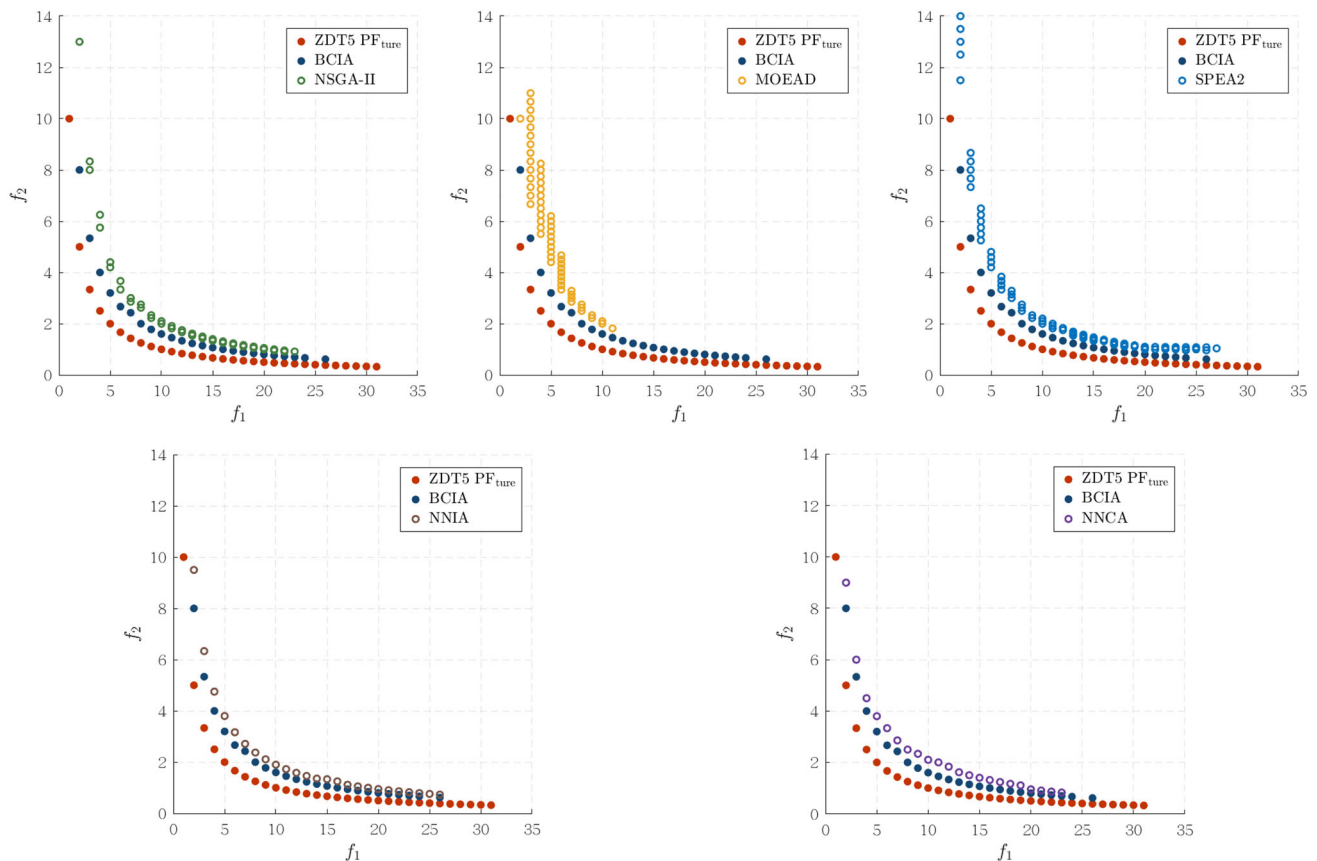
## Experimental results on MCOTAPs

In the previous subsection, we tested the BCIA's performance on benchmark MCOPs and obtained satisfactory results. Next the MCOTAP problem will be instantiated to compare the performance of BCIA with SPEA2, MOEA/D, NNIA, NNCA, H-NSGAI-VNS [21] and C-MOPSO [14].

Section 2.2 of this paper summarizes the attributes and symbols of UAVs and targets in MCOTAP (Table 1). Now we assign real and reliable values to these attributes, which are summarized in Table 4. Among them, the attributes of the UAV refer to the long-endurance reconnaissance UAV YS-20, which has participated in industry exhibitions and accepted public reports for many times [37]. The constraints of MCOTAP are uniformly set as hard constraints, that is, all solutions that do not satisfy the constraints are regarded as infeasible solutions and discarded during the evolution. The objective functions set in this experiment is consistent with

**Table 3** Comparison of HV (mean) results on benchmark MCOPs

| Problems       | N   | M | D   | NSGA-II     | MOEA/D      | SPEA2       | NNIA        | NNCA                | BCIA      |
|----------------|-----|---|-----|-------------|-------------|-------------|-------------|---------------------|-----------|
| ZDT5           | 500 | 2 | 80  | 7.5911E-1 - | 7.2752E-1 - | 7.5352E-1 - | 7.9604E-1 - | 7.8550E-1 -         | 8.0925E-1 |
| mQAP           | 500 | 2 | 10  | 6.7287E-1 - | 6.6255E-1 - | 6.7237E-1 - | 6.7080E-1 - | 6.7023E-1 -         | 6.7457E-1 |
| MOTSP          | 500 | 2 | 30  | 6.2373E-1 - | 7.0649E-1 - | 6.2568E-1 - | 7.1662E-1 - | 8.6865E-1 $\approx$ | 8.6794E-1 |
| MONRP          | 500 | 2 | 100 | 6.1226E-1 - | 1.8845E-1 - | 6.2204E-1 - | 6.8363E-1 + | 6.1696E-1 -         | 6.4305E-1 |
| +/-/ $\approx$ |     |   |     | 0/4/0       | 0/4/0       | 0/4/0       | 1/3/0       | 0/3/1               |           |

**Fig. 10** Pareto Fronts obtained by BCIA and other compared algorithms on ZDT5 benchmark problem

the general objective functions of MCOTAP defined in Sect. 2.2, which is to minimize the total completion time of the mission and the total flight range of the multi-UAV system.

In order to fully verify the performance of BCIA under different scales of problems and different distribution forms of targets, we generate two sets of MCOTAP test cases. The task targets in the two test sets follow uniform distribution and random distribution, respectively. Each set contains three task allocation problems with progressively larger problem scales. The coordinates in the first test set can be divided into five clusters with close distances around the UAV base, and the targets within the cluster are relatively evenly distributed. Among them, TestCase1-1 (TC1-1) corresponds to

the MCOTAP of 3 UAVs with 15 targets, TC1-2 corresponds to the MCOTAP of 4 UAVs with 20 targets, and TC1-3 corresponds to the MCOTAP of 5 UAVs with 30 targets. The coordinates of the target points in the second test set are randomly distributed in three progressively larger task areas. The targets in TC2-1, TC2-2 and TC2-3 are randomly generated in three task areas  $(x, y) \in [0, 30]$ ,  $(x, y) \in [0, 50]$  and  $(x, y) \in [0, 80]$ , respectively. Similarly, the three examples correspond to three scales U3-T15, U4-T20 and U5-T30, respectively. The target coordinates and the required reconnaissance time in the test cases are shown in Tables 1, 2, 3, 5 in Supplementary Material—I.

**Table 4** Attributes of UAVs and targets after instantiation

| Attributes | Value                                          |
|------------|------------------------------------------------|
| $N$        | 5                                              |
| $Vol$      | 36 km/h                                        |
| $Time$     | 5.5 h                                          |
| $S$        | (40,30)                                        |
| $N_T$      | 30                                             |
| $TP_j$     | Supplementary Material—I –Tablea 1 to Tablea 6 |
| $ToR_j$    | Supplementary Material—I -Tablea 1 to Tablea 6 |

The parameter Settings and evolutionary operators of each algorithm have been explained in Sect. "Experimental setup". This experiment also uses HV as the performance evaluation index. The accuracy of the HV is heavily influenced by the reference point. In general, benchmark problems are generally given reference point when computing HV. However, custom problems derived from real-world applications require us to calculate and select reference points. The following describes the calculate and select method taken in this experiment. We first mix all of the solutions obtained by the considered algorithms, and then calculate the non-dominated solutions of the mixed solution set. Finally, the vector formed by the maximum value in each dimension of the non-dominated solution set is used as the reference point for calculating HV [38].

The numerical experimental results of BCIA and other five comparison algorithms on the MCOTAP are summarized in Tables 6, 7, where HV and Time is the average value obtained from 30 independent experiment, Average Optimal Solution ( $f_1$ ,  $f_2$ ) is the average of the Pareto optimal solution set with the median HV value in 30 runs. The optimal values of each column have been marked by orange shading.

On all test cases, the HV of BCIA is better than all other comparison algorithms, which indicates that the comprehensive performance of BCIA is better when solving MCOTAPs. To demonstrate the superiority of BCIA more distinctly, we present the HV data in the form of box plots in Fig. 11. The position of the boxes reveals that, for each test problem, the HV data of BCIA is, on the whole, concentrated at higher positions. This indicates that BCIA performed more stably in the 30 independent runs, and could almost find a superior Pareto Front in each experiment. Although NNIA and NNCA achieved similar HV maximum values to BCIA on the TC1-1 and TC2-1 problems, their variances were larger, and their HV minimum values were inferior to those of BCIA. That is to say, they were unable to perform well stably in each run, and their reliability was lower than that of BCIA. On the TC1-3 problem, NNIA seemed to perform better than BCIA. However, when considering the outliers of the two boxes, BCIA were at higher positions above the upper quartile, while NNIA were concentrated at the bottom, far from the lower quartile. This suggests that BCIA has a higher algorithmic upper limit, while NNIA has poorer reliability and stability. In conclusion, BCIA achieved better performance on all six test problems, with excellent stability, reliability, and robustness.

In terms of Average Optimal Solution, BCIA achieves the best performance on four of the six test problems, which are TC1-2, TC2-1, TC2-2 and TC2-3. On TC1-1, the average optimal solutions obtained by all algorithms are close. This shows that the advantages and disadvantages of algorithms cannot be well demonstrated in CMTAPs with small scale and uniform distribution of targets. On TC1-3, BCIA significantly outperforms SPEA2, NNCA and H-NSGAI-VNS. However, when BCIA is compared with MOEA/D, NNIA and CMOPSO, their average optimal solutions do not

**Table 5** The average of the Pareto optimal solution set with the median HV value in 30 independent experiments

| Algorithms  | TC1-1                    | TC1-2                    | TC1-3                    | TC2-1                    | TC2-2                    | TC2-3                    |
|-------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| MOEA/D      | 4.7134e-1<br>(3.54e-2) - | 4.1905e-1<br>(3.50e-2) - | 2.9781e-1<br>(3.59e-2) - | 6.6551e-1<br>(3.34e-2) - | 5.1588e-1<br>(3.51e-2) - | 2.3777e-1<br>(3.52e-2) - |
| SPEA2       | 4.8869e-1<br>(7.85e-3) - | 4.0487e-1<br>(1.51e-2) - | 1.9341e-1<br>(1.34e-2) - | 6.8732e-1<br>(4.45e-3) - | 5.2026e-1<br>(8.32e-3) - | 2.1060e-1<br>(9.94e-3) - |
| NNIA        | 5.1173e-1<br>(1.26e-2) - | 4.6373e-1<br>(1.94e-2) - | 3.1829e-1<br>(7.79e-2) ≈ | 7.0378e-1<br>(9.31e-3) - | 5.6966e-1<br>(1.95e-2) - | 2.7451e-1<br>(3.18e-2) - |
| NNCA        | 5.1941e-1<br>(8.42e-3) ≈ | 4.7449e-1<br>(2.02e-2) - | 2.9020e-1<br>(3.38e-2) - | 7.0542e-1<br>(5.65e-3) - | 5.6765e-1<br>(1.73e-2) - | 2.9781e-1<br>(2.76e-2) - |
| H-NSGAI-VNS | 4.8962e-1<br>(8.86e-3) - | 4.0279e-1<br>(1.47e-2) - | 1.9462e-1<br>(1.27e-2) - | 6.8738e-1<br>(4.30e-3) - | 5.1692e-1<br>(9.07e-3) - | 2.1533e-1<br>(1.37e-2) - |
| C-MOPSO-CD  | 4.5485e-1<br>(1.18e-2) - | 4.0002e-1<br>(1.50e-2) - | 1.9452e-1<br>(2.79e-2) - | 6.7167e-1<br>(1.05e-2) - | 5.1730e-1<br>(8.13e-3) - | 2.0690e-1<br>(1.06e-2) - |
| BCIA        | 5.2169e-1<br>(6.21e-3)   | 4.8827e-1<br>(1.50e-2)   | 3.3686e-1<br>(2.82e-2)   | 7.0781e-1<br>(3.90e-3)   | 5.8340e-1<br>(1.88e-2)   | 3.3134e-1<br>(2.45e-2)   |

**Table 6** The average (standard deviation) of the HVs obtained from 30 independent experiments on TC1-1 to TC2-3

| Algorithms   | TC1-1              | TC1-2              | TC1-3               | TC2-1              | TC2-2              | TC2-3               |
|--------------|--------------------|--------------------|---------------------|--------------------|--------------------|---------------------|
| MOEA/D       | (332.7376, 9.2219) | (470.9085, 8.9767) | (676.9548, 10.8836) | (142.1957, 5.4854) | (331.1277, 8.1435) | (768.4236, 11.3310) |
| SPEA2        | (397.5923, 8.5418) | (565.6736, 8.6176) | (919.3724, 15.1481) | (159.3308, 6.1882) | (395.8154, 6.9167) | (905.5471, 11.4757) |
| NNIA         | (372.9787, 7.4023) | (469.0923, 8.7850) | (670.0968, 10.2682) | (137.3145, 5.1232) | (345.5444, 6.3736) | (770.8392, 9.6371)  |
| NNCA         | (343.4907, 8.7410) | (446.8773, 8.8493) | (782.5825, 9.6918)  | (138.9850, 5.0430) | (334.9118, 6.5268) | (747.9482, 10.2026) |
| H-NSGAIL-VNS | (386.9834, 8.2814) | (553.1852, 8.3994) | (933.1528, 11.7283) | (154.9625, 6.2876) | (402.6202, 7.4427) | (895.3221, 11.2885) |
| C-MOPSO-CD   | (428.6077, 8.6207) | (548.1391, 8.5812) | (686.9433, 8.8156)  | (173.1715, 7.3601) | (398.2448, 6.7974) | (967.6936, 10.8422) |
| BCIA         | (361.6169, 7.4850) | (439.3744, 7.4913) | (684.8389, 9.1680)  | (135.7237, 5.0983) | (289.9179, 6.1988) | (681.1577, 8.9397)  |

**Table 7** The average of the run time obtained from 30 independent experiments

| Algorithms   | TC1-1   | TC1-2   | TC1-3    | TC2-1   | TC2-2   | TC2-3    |
|--------------|---------|---------|----------|---------|---------|----------|
| MOEA/D       | 6.5334s | 7.5122s | 9.0181s  | 6.6731s | 7.5379s | 8.7040s  |
| SPEA2        | 5.0924s | 5.7061s | 7.0284s  | 5.0769s | 5.7343s | 6.6166s  |
| NNIA         | 2.6036s | 1.7839s | 4.4456s  | 2.6070s | 3.2948s | 4.2118s  |
| NNCA         | 2.8447s | 3.5951s | 4.9842s  | 2.8619s | 3.5467s | 4.7109s  |
| H-NSGAIL-VNS | 6.1015s | 8.1009s | 13.6880s | 6.0660s | 7.3561s | 12.0910s |
| C-MOPSO-CD   | 2.4972s | 3.3626s | 4.9694s  | 2.5788s | 3.3644s | 4.4128s  |
| BCIA         | 3.4648s | 3.8288s | 5.1037s  | 3.6251s | 3.7668s | 4.9188s  |

dominate each other. If  $f_1$  and  $f_2$  are split for analysis, the disadvantage of BCIA on  $f_1$  is smaller than its advantage on  $f_2$ . In other words, the solution obtained by BCIA can better balance the two objectives. In terms of average solving time, BCIA is significantly faster than MOEA/D, SPEA2 and H-NSGAIL-VNS but slower than NNIA, NNCA and C-MOPSO. The reason why H-NSGAIL-VNS took the longest time on all test cases is that it introduced variable neighborhood search. Based on the principles of the algorithms and the quality of the solutions, it is found that the reason for the fast solving-speed of NNIA, NNCA and C-MOPSO is falling into local optimums. This just confirms the advanced nature of BCIA. Because the adaptive coevolution mechanism proposed in this paper require the subpopulations to adaptively choose different evolutionary strategies according to the evolutionary stage, and can take into account both global search and local search, so as to ensure that BCIA will not converge to the suboptimal solutions.

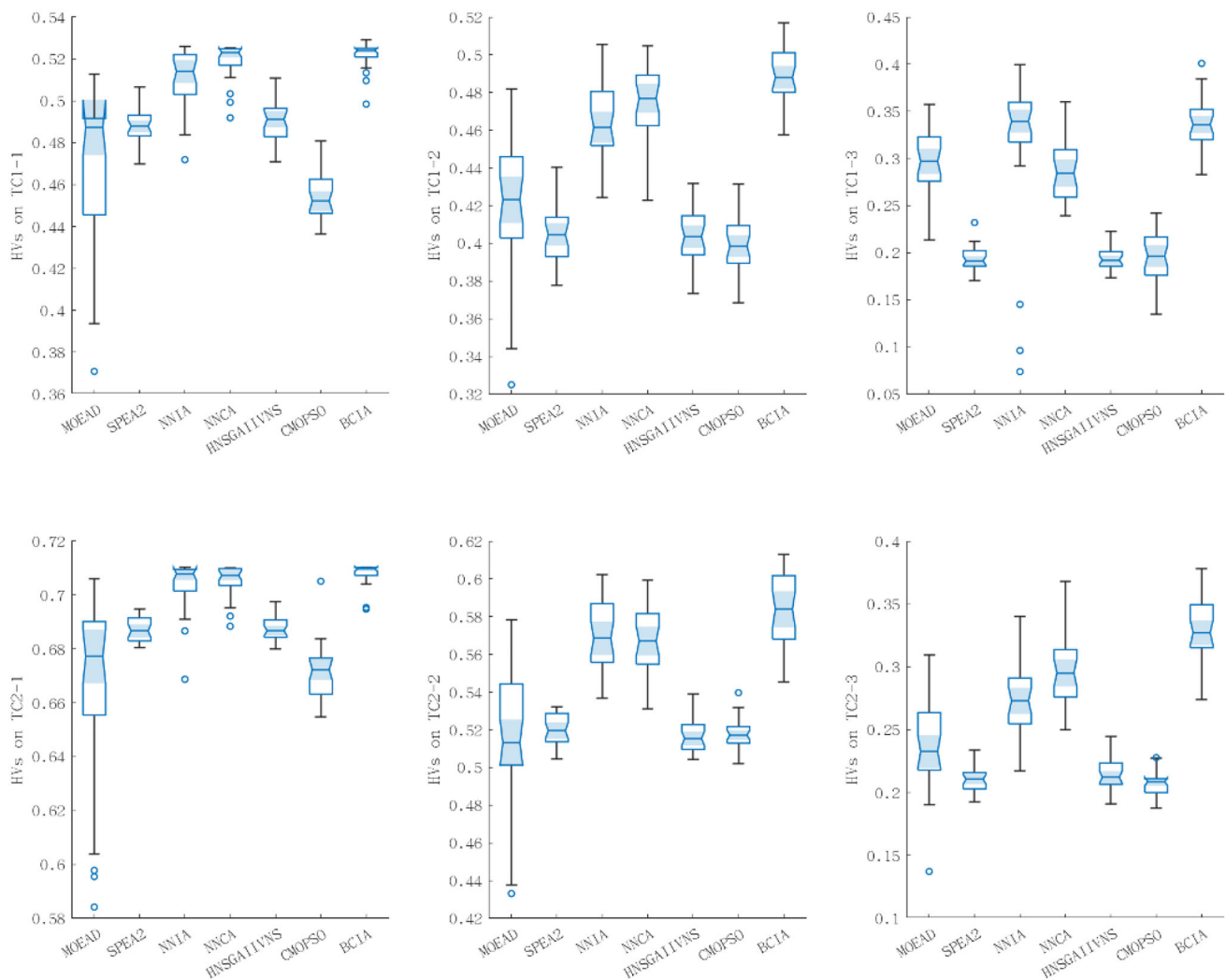
In addition, with the expansion of the problem scale, the advantages of BCIA become more obvious. Taking all the

above analyses together, we can conclude that the comprehensive performance of BCIA on the MCOTAP test set is better than these selected state-of-the-art MOEAs.

### Ablation study

To verify the effectiveness of the proposed mechanisms and operators in BCIA, we take the control variates method to conduct ablation experiments. Each variant differs from the original BCIA in only one mechanism. The variant algorithms are as follows. Supplementary Material—II describes the evolution process of the three variants and their differences with BCIA detailedly.

(1) Variant 1: There is only one population evolutionary mechanism. It only retained the elite population and the adaptive cooperative evolution within the elite population. The Variant1 is used to investigate the improvement of the algorithm ability by the Bi-subpopulation Coevolution mechanism. (2) Variant 2: The Adaptive Evolutionary mechanism is abandoned. When the two subpopulations evolve, they do not determine the evolutionary stage, and randomly adopt strategies in the evolutionary strategy pool. (3) Variant 3:



**Fig. 11** Box-plot of HVs obtained by algorithms on all MCOTAP test cases

There is only a single evolutionary strategy. The evolution of the two subpopulations only uses ES1 as the evolutionary strategy.

The results of the ablation experiment are presented in Table 8. From the contrastive results of the mean and standard deviation of HV calculated from 30 independent runs, it can be observed that BCIA outperforms all other variant algorithms in all six test problems. However, the results of the non-parametric Wilcoxon's rank-sum two-sided comparison test at the 0.05 significance level are close, indicating that the comparative advantage of BCIA in HV is not significant. Therefore, we further collated the data obtained by all algorithms in the 30 independent runs. On the one hand, we calculated the Average Optimal Solution (see Table 8); on the other hand, we visualized the experimental results corresponding to the median HV, that is, the obtained non-dominated solutions (see Table 9). Through the comparison in Table 8 and Fig. 12, it can be seen that BCIA and the variant

algorithms perform similarly on TC1-1 and TC2-1 with small problem scales. But as the problem scale expands, the advantage of BCIA becomes increasingly evident. On TC1-3 and TC2-3 with the largest problem scale, the solutions obtained by BCIA completely dominate other solutions. Therefore, the mechanisms and operators proposed in BCIA can effectively enhance the algorithm performance, especially in MCOTAP with large problem scales.

## Conclusions and future work

In this paper, we studied the multi-UAV task allocation problem from the perspective of multi-objective combinatorial optimization, and proposed a general MCOTAP model. Then, a Bi-subpopulation Coevolutionary Immune Algorithm (BCIA) is proposed to solve the MCOTAP. Compared with other MOIAs, BCIA introduces a bi-subpopulation

**Table 8** The numerical experimental results of ablation experiments

| Algorithms           |          | TC1-1                            | TC1-2                            | TC1-3                            | TC2-1                            | TC2-2                            | TC2-3                            |
|----------------------|----------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| HV:<br>Mean<br>(Std) | Variant1 | 5.2141e-1<br>(6.26e-3) $\approx$ | 4.8722e-1<br>(2.01e-2) $\approx$ | 3.1899e-1<br>(4.09e-2) -         | 7.0777e-1<br>(4.57e-3) $\approx$ | 5.7871e-1<br>(1.92e-2) $\approx$ | 3.2763e-1<br>(3.10e-2) $\approx$ |
|                      | Variant2 | 5.1829e-1<br>(8.69e-3) $\approx$ | 4.8172e-1<br>(2.74e-2) $\approx$ | 3.1306e-1<br>(3.31e-2) -         | 7.0767e-1<br>(4.50e-3) $\approx$ | 5.7913e-1<br>(1.67e-2) $\approx$ | 3.1970e-1<br>(1.94e-2) $\approx$ |
|                      | Variant3 | 5.2053e-1<br>(7.11e-3) $\approx$ | 4.8525e-1<br>(1.73e-2) $\approx$ | 3.3335e-1<br>(3.87e-2) $\approx$ | 7.0774e-1<br>(4.28e-3) $\approx$ | 5.7890e-1<br>(1.73e-2) $\approx$ | 3.2830e-1<br>(2.60e-2) $\approx$ |
|                      | BCIA     | 5.2169e-1<br>(6.21e-3)           | 4.8827e-1<br>(1.50e-2)           | 3.3686e-1<br>(2.82e-2)           | 7.0781e-1<br>(3.90e-3)           | 5.8340e-1<br>(1.88e-2)           | 3.3134e-1<br>(2.45e-2)           |
|                      | Variant1 | (361.6169,<br>7.4850)            | (417.2713,<br>9.0567)            | (706.9744,<br>9.4158)            | (138.3236,<br>5.1794)            | (337.6221,<br>6.8494)            | (697.9357,<br>9.0272)            |
| Average              | Variant2 | (347.4605,<br>7.7470)            | (441.4612,<br>8.2929)            | (711.1029,<br>9.9936)            | (136.9456,<br>5.0904)            | (305.8787,<br>6.9922)            | (695.9065,<br>10.4614)           |
| Optimal              | Variant3 | (356.1064,<br>7.6514)            | (444.5245,<br>7.1601)            | (699.7591,<br>9.9732)            | (135.5531,<br>5.0904)            | (337.6950,<br>6.5347)            | (696.0753,<br>9.3909)            |
| Solution<br>(f1, f2) | BCIA     | (361.6169,<br>7.4850)            | (427.8622,<br>6.9486)            | (663.0585,<br>8.8655)            | (135.7237,<br>5.0983)            | (289.9179,<br>6.1988)            | (681.1577,<br>8.9397)            |

coevolution mechanism and an adaptive strategy selection mechanism. The coevolution between the elite antibody population and the common antibody population improves population diversity and avoids the inherent disadvantages of immune algorithms that tend to converge prematurely to local optimums. Based on our improvement of the existing crossover operators, an evolutionary strategy pool is formed, so that the genetic selectivity in evolution can be obtained while adapting to the MCOPs. The adaptive strategy selection mechanism selects different strategies from the evolutionary strategy pool according to different stages of population evolution, so that BCIA focuses more on population diversity in the early stage and algorithmic convergence in the later stage. In the final experimental part, we compared BCIA with several other advanced MOEAs and MOIAs in the benchmark MCOPs. Furthermore, we carry out algorithm verification

and comparative analysis on two MCOTAP test sets. Experimental results show that compared with other algorithms, BCIA shows better multi-objective combination optimization performance and targeted solution ability of multi-UAV task allocation problem. Finally, we demonstrate the effectiveness of the proposed mechanisms and operators in BCIA through ablation experiments.

In the next research, we will continue to explore the multi-UAV task assignment problem, such as solving the many complex constraints in the real world. Our goal has always been to practically utilize multi-objective combinatorial optimization methods to solve real-world engineering optimization problems.

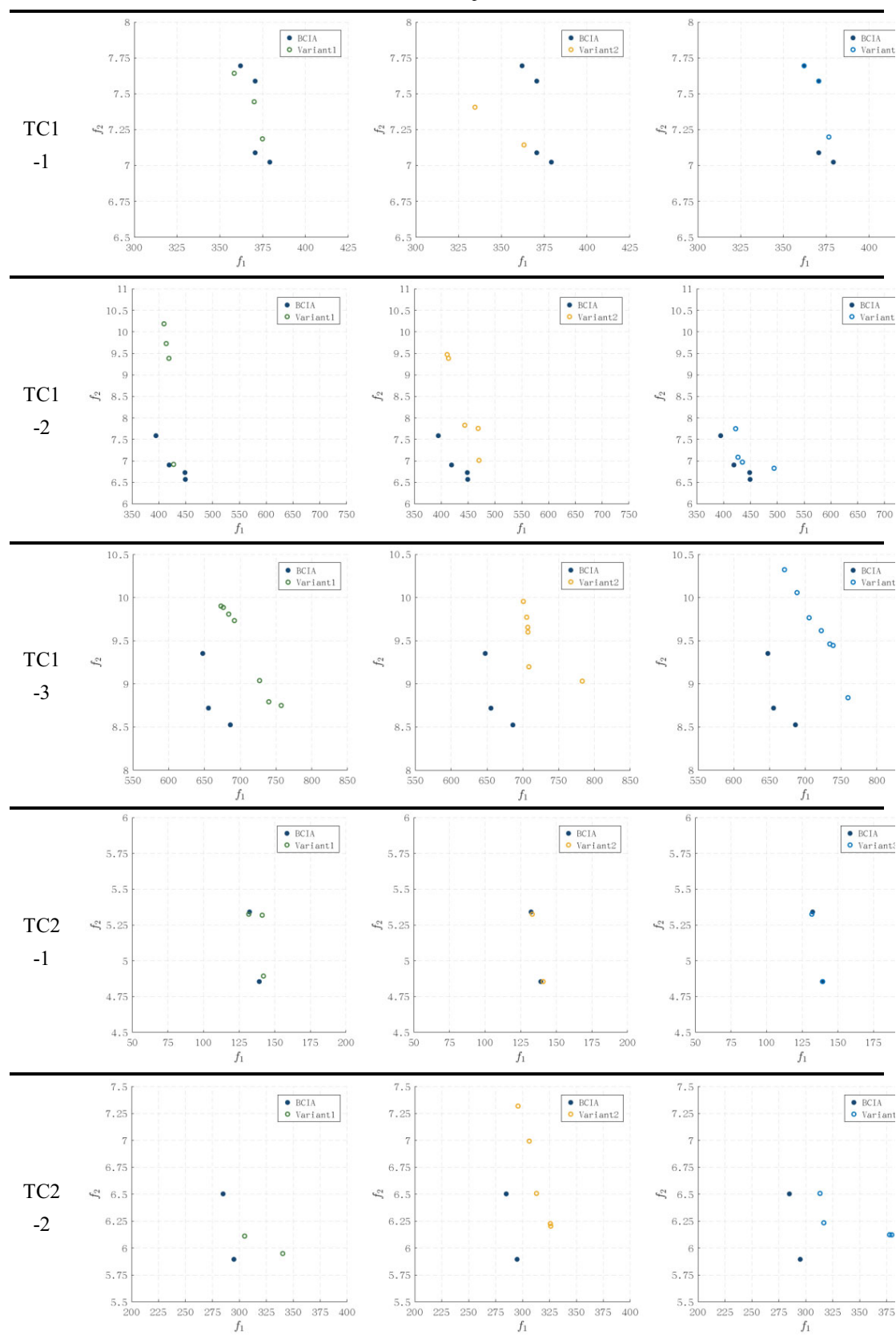
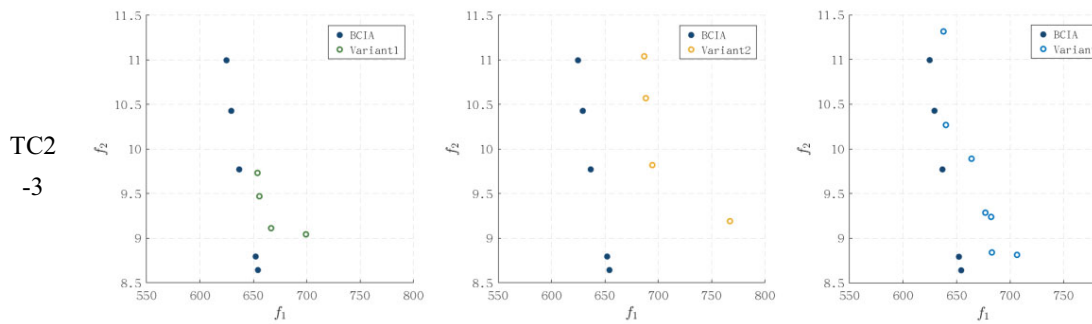
**Table 9** The non-dominated solutions obtained from ablation experiments with median HV

Table 9 (continued)



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